

SkillNav: A Multi-Agent Blackboard for Hallucination-Resistant Zero-Shot Object Navigation

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Abstract

LLM-based zero-shot object-goal navigation requires agents to retain visual, spatial, and motion evidence over long horizons. When this evidence is serialized into a growing trajectory prompt, earlier observations can become difficult to use amid redundant or conflicting context. We present SkillNav, a role-specialized multi-agent architecture that moves the navigation history needed for later decisions out of the prompt and into a shared spatial blackboard. Built on a sparse Voronoi–Frontier graph, the blackboard stores node-level evidence in role-specific fields maintained by Safe, Memory, and Exploration agents. Independent safety and memory gates reduce the influence of repeated motion failures and rejected target hypotheses, while the Exploration agent reads only a compact top- K snapshot. Their updates are combined through a unified node-value function, preserving relevant evidence without replaying the full trajectory. On the HM3D validation split, SkillNav achieves **71.9 SR / 34.0 SPL** with **22.5** calls and ~ 11.9 K tokens per episode for its high-level LLM, outperforming RATE-Nav, the strongest zero-shot baseline in our comparison, by 4.1 SR and 2.7 SPL. Ablations on a matched 200-episode subset show that removing either Safe or Memory reduces both metrics.

1 Introduction

Embodied object-goal navigation (ObjectNav) asks an agent to locate an instance of a target category in an unseen environment using egocentric observations and odometry (Chaplot et al., 2020; Savva et al., 2019). In the *zero-shot* setting (Majumdar et al., 2022), recent systems combine frozen vision-language models (VLMs) with metric maps, frontier representations, or topological graphs, sometimes using a language model for high-level decisions (Yokoyama et al., 2024; Yu et al., 2023; Wu

et al., 2024a; Zhang et al., 2024). Over an episode, the agent accumulates observations, candidate targets, visited regions, and action outcomes. For LLM-based controllers that repeatedly append this history to the prompt, the input may contain redundant or conflicting evidence. Long-context models also use information unevenly across prompt positions, a behavior known as *lost-in-the-middle* (Liu et al., 2024a). A separate problem is that a false-positive target hypothesis may persist in the decision state and bias later navigation.

A spatial map organizes where evidence is observed, but does not by itself define how different modules should update and combine that evidence. We present SkillNav (fig. 1), which uses a sparse Voronoi–Frontier graph as a shared spatial blackboard. Each node stores visual-semantic evidence and two multiplicative gates. The geometry-based Safe agent updates $\mu_{\text{safe}}(v)$ from viability and motion failures. The VLM-based Memory agent updates $\mu_{\text{mem}}(v)$ through multi-frame verification of target hypotheses. The LLM-based Exploration agent reads a compact blackboard snapshot and returns a navigation phase and $(w_{\text{SR}}, w_{\text{IG}})$ fusion weights. The agents write to separate channels, which are combined only by the node-value function. Rejected target hypotheses and motion failures can then inform later decisions without replaying their supporting observations or action history.

On the HM3D validation split, SkillNav achieves **71.9 SR / 34.0 SPL**, using an average of **22.5** calls and ~ 11.9 K tokens per episode for the high-level LLM. It exceeds RATE-Nav, the strongest zero-shot baseline in our comparison, by 4.1 SR and 2.7 SPL. Removing either Safe or Memory reduces both metrics on a matched 200-episode subset.

Contributions. (1) A Voronoi–Frontier blackboard that stores navigation evidence at sparse spatial nodes (§3.2). (2) Separate update channels for safety, memory, and exploration, combined by a

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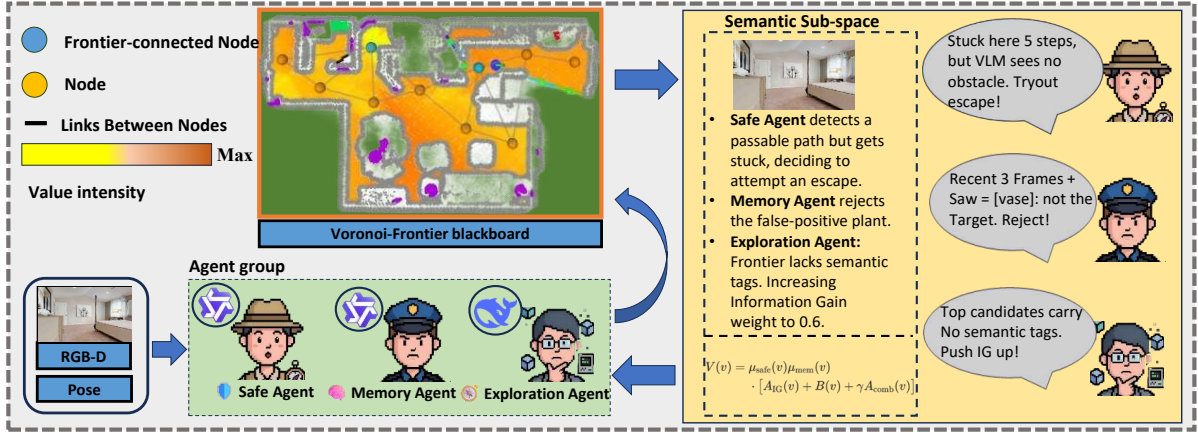


Figure 1: SkillNav grounds high-level LLM decisions in a shared Voronoi-Frontier blackboard. **Left:** RGB-D observations and pose update a sparse graph that anchors reachable frontiers. **Right:** Safe, Memory, and Exploration update separate fields of the blackboard. Their outputs are combined by a unified node-value function for frontier selection.

unified node-value function (§3.3). (3) An HM3D evaluation showing gains in SR and SPL over the strongest zero-shot baseline in our comparison, together with measured high-level LLM usage (§4).

2 Related Work

Zero-shot Object Navigation. Frontier-based ZSON systems score each frontier cell with a single VLM signal and pick the best at every step (VLFM, L3MVN, ESC (Yokoyama et al., 2024; Yu et al., 2023; Zhou et al., 2023)). TriHelper (Zhang et al., 2024) adds three helper modules sharing one planner; ImagineNav (Zhao et al., 2025) prompts a VLM to imagine future views; RATE-Nav (Li et al., 2025) adds region-aware termination. Topology methods reason at the node level instead: VoroNav (Wu et al., 2024a) runs an LLM at every Voronoi waypoint, SG-Nav (Yin et al., 2024) prompts over a 3D scene graph, CogNav (Cao et al., 2025) casts the episode as a 5-state LLM-driven process. All of them route every decision through one LM with a shared, monotonically-growing context.

Grounding and hallucination. Grounding has been studied extensively for text and knowledge bases (Lewis et al., 2020), but these works ground the LM in *retrieved text*, not in a dynamically-built spatial state. Long-context LMs lose middle-context information (Liu et al., 2024a); LLM-based embodied agents compound this into trajectory-level hallucination (Chakraborty et al., 2025; Lin et al., 2025). Common mitigations (self-revision, RAG over prior outputs) still ask the LM to cor-

rect its own past assertions; SkillNav suppresses false-positive evidence *below* the LM, in the blackboard’s algebra.

Multi-agent LLM architectures. Recent agentic systems (Wu et al., 2024b; Park et al., 2023) coordinate via free-form dialogue, recreating the long-context problem at the inter-agent level. SkillNav inherits the *blackboard* pattern from classical problem solving (Erman et al., 1980): agents write to disjoint channels, never to each other’s prompts. Combined with the options framework (Sutton et al., 1999) this yields strict context isolation.

3 Method

3.1 Overview

At time t , SkillNav receives an RGB-D observation I_t , the agent pose p_t , and a target category c . The mapping module updates the occupancy map, frontier set, and sparse Voronoi graph, which together form the spatial blackboard $\mathcal{B}_t$. Safe and Memory update node-level gates in response to motion and verification events. At scheduled high-level updates, Exploration reads a compact summary of $\mathcal{B}_t$ and returns a navigation phase and the SR/IG fusion weights. At each node-selection tick, SkillNav combines geometric information gain, connectivity, visual-semantic scores, and the two gates into a node value $V(v)$. The highest-scoring reachable node is passed to the low-level planner. Because the blackboard persists across ticks, rejected target hypotheses and motion failures remain available to later decisions without adding the corresponding observations and actions to the Exploration prompt.

3.2 The Voronoi–Frontier Blackboard

As illustrated in fig. 1, the blackboard compresses a dense map into a small set of navigation choices. A *frontier* is a boundary between explored free space and unknown space. From the current occupancy map, we extract a sparse Voronoi graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over explored free space (Lau et al., 2010). Nodes are degree- ≥ 3 junctions and edges are traversable connections between them. Each frontier $f \in \mathcal{F}$ is assigned to its nearest reachable node by an A*-aware map $\pi : \mathcal{F} \rightarrow \mathcal{V} \cup \{\perp\}$. We write $\mathcal{F}_v = \pi^{-1}(v)$ for the frontiers anchored to node v . Frontiers that cannot be assigned reliably ($\pi(f) = \perp$) are kept for a fallback policy. Thus, a node represents both a navigable waypoint and the unexplored regions reachable through it. Exact extraction and anchoring rules are given in App. E.

For a node v at grid location x_v , the blackboard stores

$$\mathcal{B}_t(v) = (x_v, \mathcal{F}_v, S(v), E(v), m(v), \mu_{\text{safe}}(v), \mu_{\text{mem}}(v)). \quad (1)$$

Here $S(v) = V_{\text{SR}}^{\text{cell}}(x_v)$ is a goal-conditioned BLIP-2 ITM score, whereas $E(v) = V_{\text{IG}}^{\text{cell}}(x_v)$ is an object-agnostic visual-explorability score. Their prompt templates are provided in App. D. The memory field $m(v)$ records a room hypothesis, observed objects, and freshness; it is included in the compact summary read by Exploration. The remaining fields are the safety and memory gates written by Safe and Memory. Exploration supplies a navigation phase and the global fusion weights ($w_{\text{SR}}, w_{\text{IG}}$).

We first compute an ungated utility $U(v)$. Its three terms measure the geometric gain of the anchored frontiers, the connectivity of the node, and its weighted visual evidence:

$$A_{\text{IG}}(v) = \sigma^{-1} \sum_{f \in \mathcal{F}_v} \text{IG}_{\text{geom}}(f), \quad (2)$$

$$B(v) = \beta \log(1 + \deg(v)), \quad (3)$$

$$A_{\text{comb}}(v) = w_{\text{SR}} S(v) + w_{\text{IG}} E(v), \quad (4)$$

$$U(v) = A_{\text{IG}}(v) + B(v) + \gamma A_{\text{comb}}(v). \quad (5)$$

The normaliser σ keeps aggregated frontier gain on a comparable scale, β controls the connectivity bonus (Wu et al., 2024a), and γ controls the contribution of visual evidence. Note that A_{IG} is *geometric* frontier gain, whereas $E(v)$ is the VLM-derived visual-explorability score.

Safe and Memory then act as independent soft gates on the whole utility:

$$V(v) = \text{clip}(\mu_{\text{safe}}(v) \mu_{\text{mem}}(v) U(v), V_{\min}, V_{\max}), \quad (6)$$

where $[V_{\min}, V_{\max}] = [0.01, 1.0]$. A node remains attractive only when it is both useful under $U(v)$ and credible under the two gates. The selected target is the highest-valued reachable node, $v^* = \arg \max_{v \in \mathcal{V}} V(v)$; orphan frontiers are handled by the fallback policy. Algorithm 1 in App. C gives the complete update and selection cycle.

3.3 Three Agents on Disjoint Channels

Table 1 summarises which channel each agent owns and how it reasons. All three agents read the same blackboard state; they never see one another’s prompts.

Memory. On each VLM verdict at v , the memory gate is updated as

$$\mu_{\text{mem}} \leftarrow \text{clip}(\gamma_e \mu_{\text{mem}}, \mu_{\min}, \mu_{\max}), \quad (7)$$

with $\gamma_e \in \{0.5, 0.7, 1.1\}$ for a multi-frame FP, a single-shot FP, and a TP respectively, and floor $\mu_{\min} = 0.1$. Three repeated REJECTs take μ_{mem} to 0.125, dropping $V(v)$ by an order of magnitude regardless of SR/IG. Repeated rejections therefore lower the node value without requiring the high-level LLM to revise an earlier decision.

Safe. Tryout escape writes

$$\mu_{\text{safe}} \leftarrow \max(\mu_{\min}, \gamma_{\text{stuck}} \mu_{\text{safe}}) \quad (8)$$

on each consecutive escape failure ($\gamma_{\text{stuck}} = 0.3$, floor 0.05); three failures would yield ≈ 0.027 before clipping and therefore reach the 0.05 floor, encoding a geometric blocker as a language-agnostic gate. The Safe agent never calls a language model.

Exploration. The Exploration agent uses DeepSeek-Chat as its high-level LLM. Every 3 s, the model receives a compact state vector (coverage, candidate count, FP count, stuck count, previous phase) and returns JSON with the next phase and a normalised ($w_{\text{SR}}, w_{\text{IG}}$). A rule-based fallback uses the same weight tiers so navigation can continue when the LLM API is unavailable (App. D).

Agent	Reads	Writes	Vocabulary	Backbone
Safe	pose, occupancy, recent trajectory	$\mu_{\text{safe}}(v)$	{wedged, stuck, tryout, ok}	pure geometry (no LM)
Memory	multi-frame RGB + per-cand. FP count	$\mu_{\text{mem}}(v)$	{CONFIRM, UNCERTAIN, REJECT}	Qwen-VL-Plus, event-trig.
Exploration	compact blackboard summary	$(w_{\text{SR}}, w_{\text{IG}})$, phase	{BROAD, DIRECTED, APPROACH, VERIFY}	DeepSeek, 3 s cadence

Table 1: The three agents of SkillNav. Each reads the shared blackboard but writes a single, named channel; vocabularies are disjoint so writes commute. Full prompts in App. D.

Method	SR $\uparrow$	SPL $\uparrow$
ESC (Zhou et al., 2023)	39.2	22.3
L3MVN (Yu et al., 2023)	48.7	23.0
VLFM (Yokoyama et al., 2024)	52.4	30.3
VoroNav (Wu et al., 2024a)	42.0	26.0
TriHelper (Zhang et al., 2024)	56.5	25.3
InstructNav (Long et al., 2024)	58.0	20.9
OpenFMNav (Kuang et al., 2024)	52.5	24.1
SG-Nav (Yin et al., 2024)	54.2	24.1
ImagineNav (Zhao et al., 2025)	53.0	23.8
ImagineNav-Oracle [†] (Zhao et al., 2025)	62.0	31.1
RATE-Nav (Li et al., 2025)	67.8	31.3
SkillNav (ours)	71.9	34.0

Table 2: HM3D val results. [†] Privileged oracle variant with real panoramic captures at candidate poses.

4 Experiments

4.1 Setup

We evaluate on the HM3D ObjectNav val split (Ramakrishnan et al., 2021) in the standard $25 \times 25 \text{ m}^2$ configuration with a 500-step horizon and 1 m success radius. SR and SPL are computed as in Savva et al. (2019). SkillNav is implemented on a Habitat-ROS bridge; its high-level LLM is DeepSeek-Chat, consulted at a fixed 3 s cadence, the Memory verifier is Qwen-VL-Plus, and both SR and IG channels score BLIP-2 ITM (Li et al., 2023) under the templates in App. D.

4.2 Main Results

Table 2 reports HM3D val SR and SPL. SkillNav reaches **71.9 SR / 34.0 SPL**, +4.1 SR / +2.7 SPL over the strongest zero-shot baseline in our comparison (RATE-Nav, ACL’25 (Li et al., 2025)). It also surpasses the privileged *ImagineNav-Oracle* variant (Zhao et al., 2025) by +9.9 SR / +2.9 SPL.

4.3 Ablation Study

Table 3 evaluates the contributions of the Safe and Memory agents and adaptive SR/IG control on the same **200** HM3D-v2 val episodes. The full model achieves 72.5 SR / 36.3 SPL on this subset.

Mem	Safe	Expl	SR $\uparrow$	SPL $\uparrow$
×	✓	✓	69.0	33.2
✓	×	✓	62.0	33.1
✓	✓	×	73.5	35.5
✓	✓	✓	72.5	36.3

Table 3: Ablations on the same 200-episode HM3D-v2 val subset. Mem/Safe ×: remove the Memory/Safe agent, respectively. Expl ×: fix SR/IG weights, retaining other Exploration functions. ✓: retain the component. All checks: full model.

Safe agent. Removing Safe causes the largest drop: 10.5 SR and 3.2 SPL. Stuck-related terminal outcomes also rise from 1 to 24. μ_{safe} retains these failures at the node level, discounting a semantically attractive waypoint after repeated failed escapes.

Memory agent. Removing the Memory agent lowers SR by 3.5 points and SPL by 3.1 points. A rejection is stored in μ_{mem} rather than left in an earlier prompt, so its penalty remains when the candidate is seen again from another view.

Exploration agent: adaptive SR/IG control. Adaptive weighting helps the agent reach likely target candidates sooner, raising SPL from 35.5 to 36.3. The same focus leaves fewer alternatives for Memory to verify and can increase commitment to an incorrect target: false-positive terminal outcomes rise from 16 to 22, while SR falls from 73.5 to 72.5.

4.4 High-Level LLM Inference Cost

Table 4 reports the measured per-episode cost of the high-level LLM in SkillNav on the HM3D val sweep. Each prompt contains a static prefix and a compact blackboard snapshot, so its size is set by topology rather than trajectory length. Per-step LLM designs such as VoroNav (Wu et al., 2024a) or L3MVN (Yu et al., 2023) consult the LM once per low-level decision; on a 500-step episode this al-

Metric (HM3D val)	Value
Calls / episode (mean)	22.5
Prompt tokens / call (mean)	527
Tokens / episode (mean)	11,885
LM call latency (mean / p_{95})	1.95 s / 2.50 s
VLM verifier calls / episode	4.54

Table 4: Measured high-level LLM cost on HM3D val. Token counts exclude the Memory VLM (4.54 calls/episode) and dual BLIP-2 ITM scoring.

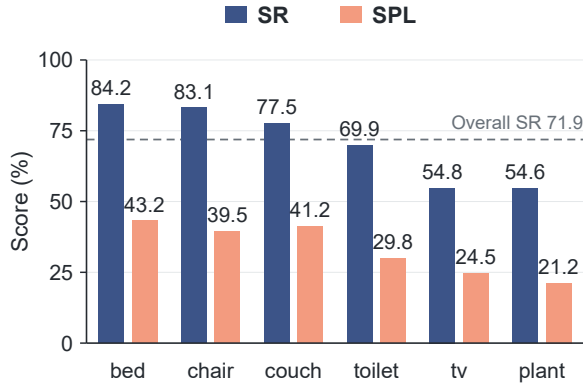


Figure 2: Per-category SR and SPL on HM3D val. The dashed line marks the overall SR (71.9).

lows up to ~ 500 calls. SkillNav’s option-boundary cadence cuts this to 22.5 calls per episode while improving SR (71.9 vs. VoroNav’s 42.0 in table 2).

Figure 2 shows over 75% SR for large furniture but about 55% for small or partially occluded targets, consistent with prior ObjectNav results (Yokoyama et al., 2024; Zhang et al., 2024). Detection remains the dominant failure mode (App. A).

4.5 Context Isolation and Hallucination Resistance

A monolithic agent serialises observations, decisions, and recovery attempts into a growing trajectory prompt. SkillNav instead stores typed fields: Memory writes its verification state and μ_{mem} , Safe writes μ_{safe} , and Exploration reads only the top- K nodes and candidates. Earlier visual and motion evidence can then affect later decisions without being replayed.

An early rejection or stuck event is thus carried forward as a value attached to a node, not as prose appended to every later prompt. At 527 tokens per call and 22.5 calls per episode (table 4), the prompt is bounded by the topology rather than trajectory length. The high-level LLM chooses from the cur-

rent state instead of replaying the trajectory. Earlier corrections therefore do not have to be recovered from the middle of a long prompt.

Channel ownership also prevents one agent from overwriting another’s evidence. A decision can be traced to a named field without reconstructing the action history. The compression is not lossless, however: restricting the snapshot to top- K entries may omit a useful distant frontier.

The full run still has 83 false-positive lock-ins (8.3%; App. A). The Memory ablation tests whether a rejection persists, not whether the VLM produces fewer false positives. Direct frame-level tests and a matched monolithic baseline are still needed.

5 Conclusion

SkillNav uses a Voronoi–Frontier blackboard to separate persistent navigation evidence from high-level choice. Safe records motion failures, Memory stores the status of target hypotheses, and Exploration sets the fusion weights and phase from a topology-bounded view. SkillNav reaches 71.9 SR / 34.0 SPL with 22.5 LM calls per HM3D val episode. On the ablation subset, removing Safe causes the largest drop, while removing Memory also lowers both metrics. Adaptive SR/IG weighting gives higher SPL but slightly lower SR than fixed weights.

Limitations

Our evaluation is limited to HM3D validation in Habitat and one VLM/LM stack; cross-simulator and cross-model generalisation remain untested. Memory updates can lag the current pose, and the fixed-size LLM summary omits graph-global structure that a graph-attention head could exploit.

Ethics Statement

SkillNav is evaluated only in HM3D simulation, with no human subjects or private data. DeepSeek and DashScope APIs are used under their standard terms. We see no direct dual-use risk beyond embodied navigation research.

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A Full Failure-Mode Taxonomy

Automatic outcome breakdown over all **281** HM3D val failures (of 71.9% success):

- **Timed-out on correct path (61.2%** of failures, 17.2% of all episodes): the agent ran out of the 500-step budget along a feasible path to the goal; combines the `stepout_feasible` (10.0%) and `stepout_fn` (7.2%) outcome tags.
- **VLM false-positive locked (29.5%** of failures, 8.3% of episodes): the agent committed to a candidate that the Memory agent’s verification eventually rejected, but not in time to recover within the step budget.
- **Genuine unreachable / no-frontier (6.0%** of failures, 1.7% of episodes): the target is behind a closed door, on another floor, or the explored region was exhausted with no remaining frontier; no purely navigational policy could succeed (`infeasible + no_frontier(_fn)`).
- **Un-escaped dead-zone (3.2%** of failures, 0.9% of episodes): the Safe agent’s μ_{safe} decay isolated a region whose re-entry was later needed (`stucking + stucking_fn`).
- **Map error / SLAM drift (<1%)**: rare in Habitat with provided GT pose; included for taxonomy completeness.
- **Other / unclassified (0%)**.

B Hyperparameters

Table 5 lists every hyperparameter introduced in §3.

C End-to-End Decision Procedure

D Prompt Templates

Semantic-relevance prompt. The SR channel uses a goal-conditioned BLIP-2 template:

“a {goal} in the room”

where {goal} is the episode’s object category (e.g. `toilet`, `potted plant`). The cosine similarity between this text and the current view is rendered into $V_{\text{SR}}^{\text{cell}}(x)$ across the agent’s projected field of view.

Symbol	Meaning	Value
<i>Frontier-anchored topology (§3.2)</i>		
λ	geom-IG weight in A_{IG}	0.5
β	degree-bonus weight	0.5
$r_{\max}$	anchor radius (A^*)	5.0 m
τ	orphan threshold (%-ile)	30
$d_{\deg}$	min node degree	3
<i>Memory agent (§3.3)</i>		
K_{ver}	verification frame window	5
γ_{fp}	per-FP penalty	0.1
λ_1	OppCost foregone weight	0.5
λ_2	OppCost FP-risk weight	1.0
<i>Safe agent (§3.3)</i>		
T_{stuck}	stuck window (ticks)	30
γ_{stuck}	per-failure μ decay	0.3
$\mu_{\min}$	multiplicative floor	0.05
<i>Exploration agent (§3.3)</i>		
Δt_{LM}	high-level LLM cadence	3.0 s
$\rho_{\text{lo}}/\rho_{\text{hi}}$	coverage tiers	0.3/0.7
K	top-K nodes/candidates	8
<i>Habitat / ObjectNav (§4.1)</i>		
horizon	step budget	500
r_{success}	success radius	1.0 m
sensor	RGB-D, HFoV	79°

Table 5: Hyperparameters used in SkillNav.

Information-gain prompt. The IG channel uses a single, object-agnostic explorability template:

“An image of a doorway, corridor, or open passage leading to an unexplored region of the building.”

The template captures the classical explorability prior that doorways, corridors, and partial openings tend to reveal large unmapped areas. We experimented with multi-template averaging (three variants emphasising *doorway*, *corridor*, *opening* respectively); the gain over single-template was within ± 0.3 SR on HM3D val, and the single-template version is what we report in the main text.

Option-boundary LLM consultation. At each option-boundary tick or stagnation event, the Exploration agent sends a structured prompt to the language-model backend (DeepSeek-Chat) following the template in table 6. Each component is bounded in size: the topology summary lists at most $K = 8$ nodes, each formatted as $\langle \text{id}, \text{visited}, \text{nearby_objects}, V(v) \rangle$ in ~ 30 tokens; the candidate summary is similar; the total prompt stays under ~ 1.3 K tokens regardless of episode length.

Algorithm 1 SkillNav decision cycle

Input: RGB-D observation I_t , pose p_t , target category c ; persistent blackboard $\mathcal{B}_{t-1}$.

Phase 1: Map and score update

- 1: Update the occupancy map and dense SR/IG score maps from (I_t, p_t, c) .
- 2: Extract $\mathcal{G}_t$ and $\mathcal{F}_t$; anchor each frontier with π_t and retain unassigned frontiers.

Phase 2: Blackboard update

- 3: Refresh $m(v)$ and the Safe/Memory gates when their respective motion or verification triggers fire.
- 4: If a high-level update is due, obtain the phase and $(w_{\text{SR}}, w_{\text{IG}})$ from Exploration using a compact summary of $\mathcal{B}_t$.

Phase 3: Node selection

- 5: For each v , compute $U(v)$ by eq. (5) and gated value $V(v)$ by eq. (6).
- 6: Set $v^* \leftarrow \arg \max_{v \in \mathcal{V}} V(v)$; invoke the orphan-frontier fallback if no valid anchored node remains.
- 7: Send v^* to the low-level planner and retain $\mathcal{B}_t$ for the next tick.

Figure 3: End-to-end decision cycle. Agent-specific update rules are given in §3.3; map and frontier implementation details are given in App. E.

E Implementation Details

Source code. Source code is available at <https://github.com/Pandakingx/bc/SkillNav>.

Habitat / ROS stack. SkillNav is implemented as a Python evaluator (`habitat_evaluation.py`) communicating with a C++ ROS planner (`exploration_node`) via standard ROS topics. The evaluator publishes RGB-D + odometry at ~ 3 Hz and the dual ITM scores at ~ 5 Hz; the planner publishes the current goal node, trajectory, and option state at ~ 1 Hz. Each episode is preceded by a hard reset of the planner process to ensure no cross-episode state leakage between scenes.

VLM serving. We host BLIP-2 ITM, Grounding-DINO (Liu et al., 2024b), and SAM (Kirillov et al., 2023) as independent GPU-backed Flask services listening on ports 12181–12184. The dual ITM is a single BLIP-2 endpoint queried twice per frame (once with the SR prompt, once with the IG prompt); we cache the visual encoder forward pass on the server side to amortise the cost.

Voronoi topology. The generalised Voronoi diagram is computed from the obstacle distance field at each map update (Lau et al., 2010). Decision nodes are extracted as Voronoi vertices with degree ≥ 3 (junctions), filtered by a minimum inter-node distance of 1.5 m and a minimum obstacle clear-

System. You are the Exploration agent for a zero-shot object-goal navigation robot. Output the next navigation phase, and optionally an override of the SR / IG fusion weights. Respond in JSON with keys: phase, w_sr, w_ig.

User. Target object: {goal}.
 Current phase: {current_option}.
 Phase history (last 5): [{o_{t-4}}, ..., {o_t}].
 Top- K topology nodes (sorted by $V(v)$), each formatted as
 {id, visited, nearby=[obj_1, ...], V}.
 Top- K target candidates (sorted by reliability), each as
 {obj, rel, fp, dist}.
 Choose the next phase from {broad-explore, directed-search, target-approach, verification}.

Table 6: Template for the Exploration agent’s LLM consultation (§3.3). Slot values are populated from the current blackboard state by the Exploration agent.

ance of 0.3 m. The graph typically contains 30–100 nodes mid-episode in HM3D scenes.

Frontier extraction and anchoring. Frontiers are extracted as 8-connected boundaries between free and unknown cells, clustered into groups of ≥ 5 cells. The anchor map π (§3.2) is computed by running a multi-source A* from all Voronoi nodes simultaneously; cells inherit the closest node’s id when their reachable distance is $\leq r_{\max} = 5.0$ m, else they enter the orphan queue.

Token instrumentation. LLM call statistics are collected by a lightweight wrapper around each backend client; for DeepSeek we use the exact prompt_tokens and completion_tokens fields from response.usage, and for Ollama the exact prompt_eval_count / eval_count fields. VLM call latency is measured at the client side as wall-clock around the Flask request. Where exact token counts are unavailable we fall back to a character-based approximation (~ 4 chars / token for ASCII, ~ 2 for CJK), and this approximation is used for $<1\%$ of the calls in our final runs.

Compute. All evaluation was performed on a single workstation with one NVIDIA GeForce RTX 4070 Ti GPU for the local VLM servers (BLIP-2 ITM, Grounding-DINO, SAM), one CPU thread per Habitat simulator instance, and the DeepSeek-Chat and DashScope (Qwen-VL-Plus) API endpoints for the high-level LLM and Memory verifier respectively. Wall-clock per HM3D episode

averaged 85.6 s (median 60.6 s) for SkillNav; the full 1000-episode HM3D val sweep completed in 23.8 h.